PALMER GROVE

# Feedback on the skills to close

Pure Mathematics 2 • WMA12/01

Sultan June 2024 and October 2024 scripts

This sheet is written from the two marked scripts. It is a Palmer Grove teaching note, not an official Pearson report, and it does not reproduce the papers. The algebra, the log laws that lead to a quadratic, the trapezium rule and the arithmetic-series sum are already scoring. The marks still being left behind sit in a short list of question types. Work down the list. Do not revise the whole of P2 at the same speed.

**Already secure, so do not spend the week here.**

Trapezium rule with a correct strip width. Remainder theorem and the simultaneous equations that follow it. Log laws, including rejecting a root that makes a log undefined. Arithmetic series, including $S_n = \frac{n}{2}\big(2a + (n-1)d\big)$. A counter-example, once a prime has been tested by division.

## Work in this order

1. **Circles, once the centre is known.** June 2024 Question 7(b) stopped after the distance. October 2024 Question 10 did not start the tangent condition, and the radius was written as $20$ before $\sqrt{20}$.
2. **The constant term of a binomial product.** October 2024 Question 5(a) was right. Part (b) dropped the $a$ in $1152a$.
3. **The period of a sequence, then a long sum.** October 2024 Question 2(a) was shown. Part (c) was treated as period $5$, and not finished. The repeating block has length $4$.
4. **A fractional power, differentiated and then rearranged.** October 2024 Question 4 used halves for thirds. June 2024 Question 9 had the right derivative, then turned $12 = 5x$ into $x = \frac{5}{12}$.
5. **A definite integral written once, in the right order.** October 2024 Question 8 reached $\frac{16}{3}$ on a page the marker could not follow. June 2024 Question 6(c)(ii) was not linked back to the trapezium estimate.
6. **One log with a root as the base, and one exponential equation left unfinished.** June 2024 Question 3(ii) and Question 10(c).

### 1. Circles: radius, then the comparison — June Q7 • October Q10

The centre is the part that is already under control. June: centre $(-4,\ 5)$ and radius $\sqrt{70}$. October: centre $(-2,\ 15)$. The radius is the square root of the number on the right of the completed square, not that number itself.

$$x^2 + y^2 + 4x - 30y + 209 = 0 \quad \Longrightarrow \quad (x+2)^2 + (y-15)^2 = 20.$$

The radius is $\sqrt{20} = 2\sqrt{5}$, not $20$.

**Two circles.** Find the distance between the centres and compare it with the sum and the difference of the radii. In June the centres are $(-4,\ 5)$ and $(5,\ -8)$, so

$$d = \sqrt{9^2 + 13^2} = \sqrt{250} = 5\sqrt{10}.$$

The radii are $\sqrt{70}$ and $\sqrt{52}$. The distance $5\sqrt{10} \approx 15.8$ is not yet a proof. Compare it with the sum:

$$\big(\sqrt{70} + \sqrt{52}\big)^2 = 122 + 2\sqrt{3640}, \qquad (5\sqrt{10})^2 = 250.$$

Since $128^2 = 16384 > 4 \times 3640 = 14560$, we have $128 > 2\sqrt{3640}$, so $d > \sqrt{70} + \sqrt{52}$. The circles are separate: they neither touch nor intersect. Stopping at $15.8$ leaves that sentence unwritten, which is where the marks went.

**A tangent from a line $y = mx + 1$.** Substitute into the circle. One quadratic in $x$ appears. A tangent meets the circle once, so the discriminant is zero:

$$b^2 - 4ac = 0.$$

That single condition is the equation in $m$. October asked for $2m^2 - 7m - 22 = 0$. It is the discriminant, not a second diagram.

## 2. Binomial expansion: which products are constant? — October Q5

The first three terms were right:

$$(2 + ax)^6 = 64 + 192ax + 240a^2x^2 + \cdots$$

The other factor is

$$\left(3 + \frac{1}{x}\right)^2 = 9 + \frac{6}{x} + \frac{1}{x^2}.$$

A constant term has power $x^0$. Three products do that, and each must keep its $a$:

$$\begin{aligned} 9 \times 64 &= 576, \\ \frac{6}{x} \times 192ax &= 1152a, \\ \frac{1}{x^2} \times 240a^2x^2 &= 240a^2. \end{aligned}$$

The $1152$ is not a constant. It is $1152a$. Set the constant term equal to $576$:

$$576 + 1152a + 240a^2 = 576 \quad \Longrightarrow \quad a(240a + 1152) = 0.$$

Since $a \neq 0$, $a = -\dfrac{1152}{240} = -\dfrac{24}{5}$.

## 3. A repeating sequence: count the block, not the return — October Q2

$u_5 = u_1$ does not make the period $5$. The terms that repeat are four long:

$$u_1 = 7, \quad u_2 = k - 7, \quad u_3 = 2k - 7, \quad u_4 = 7 - k,$$

and then $u_5 = 7$ starts the block again. One block sums to $2k$. The first four terms sum to $30$, so $k = 15$, and the block is

$$7,\ 8,\ 23,\ -8.$$

For $150$ terms, divide by the period:

$$150 = 4 \times 37 + 2.$$

There are $37$ full blocks and then the first two terms of the next one.

$$\sum_{r=1}^{150} u_r = 37 \times 30 + 7 + 8 = 1125.$$

Write the remainder before you add. The mark is the remainder, not a longer copy of part (a).

## 4. Fractional powers — October Q4 • June Q9

Read the power that is printed. $x^{1/3}$ is not $x^{1/2}$.

$$y = 4x^{1/3} + 9x^{-1/3} + 3, \qquad \frac{dy}{dx} = \frac{4}{3}x^{-2/3} - 3x^{-4/3}.$$

At a stationary point the derivative is zero. Multiply through by $x^{4/3}$, which is safe for $x > 0$:

$$\frac{4}{3}x^{2/3} = 3 \quad \Longrightarrow \quad x^{2/3} = \frac{9}{4} \quad \Longrightarrow \quad x = \left(\frac{3}{2}\right)^3 = \frac{27}{8}.$$

The second derivative, and the range where the curve is decreasing, are then about this value of $x$. A correct second-derivative test on the wrong function cannot rescue part (a).

In June the derivative itself was right:

$$y = 8x^{3/2} - 2x^{5/2}, \qquad \frac{dy}{dx} = 12x^{1/2} - 5x^{3/2}.$$

$$12x^{1/2} = 5x^{3/2}.$$

Divide by $x^{1/2}$, not by $12$ on the wrong side:

$$12 = 5x, \qquad x = \frac{12}{5}.$$

The slip was $x = \dfrac{5}{12}$. Say the sentence "divide by the lowest power of $x$" before you move any coefficient.

## 5. Definite integrals the marker can read — October Q8 • June Q6

On $1 \le x \le 3$ the upper curve is $13 - 9x^{-2}$, not $x^2 + 3$. Write one integral, then one antiderivative, then the two substitutions.

$$\int_1^3 \left[\left(13 - \frac{9}{x^2}\right) - (x^2 + 3)\right] dx = \int_1^3 \left(10 - x^2 - 9x^{-2}\right) dx$$

$$= \left[10x - \frac{x^3}{3} + \frac{9}{x}\right]_1^3 = 24 - \frac{56}{3} = \frac{16}{3}.$$

$\dfrac{16}{3}$ was on the script. The marks went because the order of the curves and the antiderivative were crossed out and rewritten on top of each other. One clean column outscores a correct number that cannot be read.

The same habit fixes June Question 6(c). You already have an estimate of $\displaystyle\int_2^{3.5} (2^x - 2x)\,dx$. Use it. Do not start again.

$$2^{x+1} = 2 \cdot 2^x,$$

so

$$\int_2^{3.5} (2^{x+1} - 4x)\,dx = 2\int_2^{3.5} 2^x\,dx - 2\left[x^2\right]_2^{3.5}.$$

And $\displaystyle\int 2^x\,dx$ is the trapezium estimate plus $\displaystyle\int 2x\,dx$. The link is the mark. A fresh, unfinished attempt at $2^x \times 2$ does not use the earlier part.

## 6. The two log questions that were left open — June Q3(ii) and Q10(c)

The ordinary log equation was fine, including the rejection of $x = 26$ because $2 - x$ must be positive. The unfinished part was the change of base. If $a > 1$,

$$\log_{\sqrt{a}}\left(a^6\right).$$

Since $a = (\sqrt{a})^2$, we have $\log_{\sqrt{a}} a = 2$, and so

$$\log_{\sqrt{a}}\left(a^6\right) = 6\log_{\sqrt{a}} a = 6 \times 2 = 12.$$

Equally, $a^6 = (\sqrt{a})^{12}$, so the logarithm is $12$ at sight. The script had the change-of-base fraction and then stopped.

For the two geometric models, the missing line is the one that takes logs. If the dormouse model is $2000(1.03)^T$ and the vole model is $ab^T$,

$$2000(1.03)^T = ab^T \quad \Longrightarrow \quad T = \frac{\ln(a/2000)}{\ln(1.03/b)}.$$

Keep more than two figures in $b$ until the last line, then round $T$ as the question asks. Part (b) had already found $a$ and $b$. Part (c) is that one division.

## Two shorter habits

**Every solution in the interval.** October Question 9(a), the identity, was shown. In part (b), $\sin\theta = (-1+\sqrt{10})/3$ gives a principal value and also $\pi$ minus that value. Then add and subtract $2\pi$, and keep every angle which puts $x$ inside $-\pi < x < \pi$. One value in that interval was not listed. Write the range for the substituted angle before you solve, so a solution is not lost at the edge.

**The last sentence of a proof.** October Question 11(i), the counter-example $n = 5$, was enough: $3^5 + 2 = 245 = 5 \times 49$. In part (ii) the two expansions were right. An integer not divisible by $3$ is $3k+1$ or $3k+2$, and both give $m^2 - 1 = 3 \times (\text{an integer})$. The written conclusion said "$m$ is not $3$". That is a different statement. Copy the claim you were asked to prove, and stop.

## Five questions to do before the next paper

These are original practice, in the same styles. The repairs are on the next page. Try them first.

1. The circle $(x-1)^2 + (y+2)^2 = 25$ and the circle $x^2 + y^2 - 8x - 6y + 16 = 0$. Find the two centres and the two radii, then decide whether the circles intersect.
2. Find the constant term in $\left(2 + \dfrac{3}{x}\right)^2 (1+ax)^4$, and the non-zero value of $a$ for which that constant term is $4$.
3. A sequence repeats every three terms: $5,\ -1,\ 4,\ 5,\ -1,\ 4,\ \ldots$ Find $\displaystyle\sum_{r=1}^{100} u_r$.
4. $y = x^{1/2} + 4x^{-1/2}$, for $x > 0$. Find the coordinates of the stationary point.
5. Solve $500(1.02)^t = 800(0.95)^t$, giving $t$ to one decimal place.

## Answers to the five questions

**1.** First centre $(1, -2)$, radius $5$. Second circle: $(x-4)^2 + (y-3)^2 = 9$, so centre $(4, 3)$, radius $3$.

$$d = \sqrt{3^2 + 5^2} = \sqrt{34} \approx 5.83.$$

The sum of the radii is $8$ and the difference is $2$. Since $2 < d < 8$, the circles intersect at two points.

**2.** $\left(2 + \dfrac{3}{x}\right)^2 = 4 + \dfrac{12}{x} + \dfrac{9}{x^2}$ and $(1 + ax)^4 = 1 + 4ax + 6a^2x^2 + \cdots$. The constant term is

$$4 \times 1 + \frac{12}{x} \times 4ax + \frac{9}{x^2} \times 6a^2x^2 = 4 + 48a + 54a^2.$$

Set this equal to $4$: $a(54a + 48) = 0$. Since $a \neq 0$, $a = -\dfrac{8}{9}$.

**3.** One block sums to $5 + (-1) + 4 = 8$. Then $100 = 3 \times 33 + 1$, so the sum is $33 \times 8 + 5 = 269$.

**4.** $\dfrac{dy}{dx} = \dfrac{1}{2}x^{-1/2} - 2x^{-3/2}$. Set this equal to zero and multiply by $x^{3/2}$:

$$\frac{1}{2}x - 2 = 0, \qquad x = 4.$$

Then $y = 2 + 4 \cdot \dfrac{1}{2} = 4$. The stationary point is $(4,\ 4)$.

**5.**

$$\left(\frac{1.02}{0.95}\right)^t = \frac{800}{500} = 1.6, \qquad t = \frac{\ln 1.6}{\ln(1.02/0.95)} \approx 6.6.$$

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